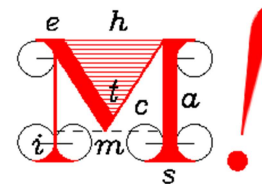


Olympiad Geometry VI: Using Coordinate Geometry in Olympiad Problems



As long as algebra and geometry proceeded along separate paths, their advance was slow and their applications limited. But when these sciences joined company they drew from each other fresh vitality and henceforth marched on at a rapid pace toward perfection.

Joseph Louis Lagrange (1736 - 1813)

1. Basic results in Coordinate Geometry

Distance, gradient and section formulae

Distance between $A(x_1, y_1)$ and $B(x_2, y_2) = AB = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

Slope of the line joining $A(x_1, y_1)$ and $B(x_2, y_2) = m_{AB} = \frac{y_1 - y_2}{x_1 - x_2}$

The point P that divides AB in the ratio $\lambda:\mu$ is $\left(\frac{\mu x_1 + \lambda x_2}{\lambda + \mu}, \frac{\mu y_1 + \lambda y_2}{\lambda + \mu} \right)$.

In particular, the midpoint of $AB = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

Angle between two lines

The acute angle θ between two given lines with slopes m_1 and m_2 is given by $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$.

It follows that two lines are parallel if and only if $m_1 = m_2$; perpendicular if and only if $m_1 m_2 = -1$.

Area of polygon

The area of the polygon with vertices $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ is $\frac{1}{2} \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_n & y_n \\ x_1 & y_1 \end{vmatrix}$.

Equation of straight Lines

Point-Slope Form $y - y_1 = m(x - x_1)$

Parametric Form $\begin{cases} x = x_1 + r \cos \theta \\ y = y_1 + r \sin \theta \end{cases}, \quad r \in \mathbf{R}$

Family of straight lines passing through the intersection of $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is $a_1x + b_1y + c_1 + \lambda(a_2x + b_2y + c_2) = 0$

□

Distance from a point to a line and angle bisector

The distance from the point (x_1, y_1) to the line $ax + by + c = 0$ is $\left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$.

The equation of the bisector of the angles between the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

Collinear Points

Three points (x_1, y_1) , (x_2, y_2) , (x_3, y_3) are collinear if and only if
$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0.$$

Proof:

The three points are collinear

$\Leftrightarrow$ There exists straight line $ax + by + c = 0$ (not all a , b , and c are zero) such that $ax_i + by_i + c = 0$ for $i = 1, 2, 3$.

$$\Leftrightarrow \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ has non-trivial solution } (a, b, c)^T \Leftrightarrow \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0.$$

Concurrent Lines

If the three straight lines
$$\begin{aligned} a_1x + b_1y + c_1 &= 0 \\ a_2x + b_2y + c_2 &= 0 \\ a_3x + b_3y + c_3 &= 0 \end{aligned}$$
 are concurrent, then
$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

Proof:

The three lines intersect at (x', y')

$$\Rightarrow (x', y', 1)^T \text{ is a non-trivial solution of the homogeneous system } \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$$\Rightarrow \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

Remark:

The converse of the above theorem is NOT true.

Consider the three lines:

$$x + 2y + 1 = 0$$

$$2x + 4y + 3 = 0$$

$$5x + 10y + 7 = 0$$

$$\text{We have } \begin{vmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 5 & 10 & 7 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 2 \end{vmatrix} = 0.$$

But the three lines are NOT concurrent (Can you see why?)

Forms of equation of circle

The standard equation of a circle with centre (a, b) and radius r is given as

$$(x - a)^2 + (y - b)^2 = r^2$$

General form of a circle is $x^2 + y^2 + 2gx + 2fy + c = 0$

with centre $(-g, -f)$ and radius $= \sqrt{g^2 + f^2 - c}$

The equation of a circle with (x_1, y_1) and (x_2, y_2) as endpoints of a diameter is

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

Tangent at a given point

The equation of the tangent to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at (x_1, y_1) is

$$x_1x + y_1y + g(x + x_1) + f(y + y_1) + c = 0$$

Family of Circles

(1) Let $C_1: x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$

$$C_2: x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$$

be two circle. Then the equation

$$x^2 + y^2 + 2g_1x + 2f_1y + c_1 + \lambda(x^2 + y^2 + 2g_2x + 2f_2y + c_2) = 0, \quad \lambda \in \mathbb{R} \setminus \{-1\}$$

represents a system of circles passing through the intersection of C_1 and C_2 .

(2) Let $C: x^2 + y^2 + 2gx + 2fy + c = 0$

$$L: px + qy + r = 0$$

The equation

$$x^2 + y^2 + 2gx + 2fy + c + \lambda(px + qy + r) = 0$$

represents a system of circles passing through the intersection of C and L .

Radical axis of two circles

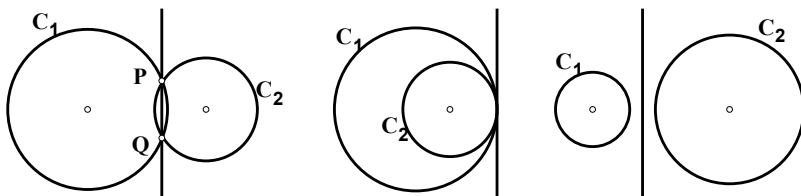
Let $C_1: x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$

$$C_2: x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$$

be two circle. Then

$$C_1 - C_2: 2(g_1 - g_2)x + 2(f_1 - f_2)y + (c_1 - c_2) = 0$$

is a straight line known as the **radical axis** of the two circles, and is the common chord when the circles intersect. In particular, if the circle touch, the radical axis is the common tangent at their point of contact.



Fact: The radical axis of two circles is perpendicular to the line joining their centres.

2. Problems

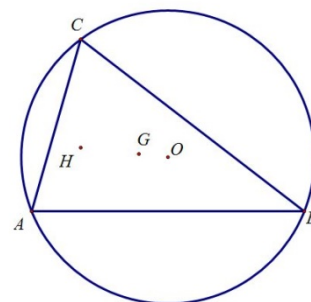
When we do a geometry problem, we should first look at the given facts and the conclusion. If all these involve intersection points, midpoints, feet of perpendiculars, parallel lines, then there is a good chance we can solve the problem by coordinate geometry. However, if they involve two or more circles, angle bisectors and areas of triangles, then sometimes it is still possible to solve the problem by choosing a good place to put the origin and the x -axis. Below we will give some examples.

Problem 1 (Criterion for orthogonality)

Let AC and BD be two lines in plane. Then $AC \perp BD$ if and only if $AB^2 + CD^2 = AD^2 + BC^2$.

Problem 2 (Euler line)

Let ABC be a non-equilateral triangle and let H , G , O be its orthocentre, centroid, and circumcentre respectively. Then the points H , G , O lie on a single line (called the **Euler line** of triangle ABC) in this order, and $HG = 2GO$.

**Problem 3**

Let AD be the altitude of $\triangle ABC$ with D on BC and H is a point on AD . The extensions of BH and CH meet AC and AB respectively at E and F . Show that AD bisects $\angle EDF$.

Problem 4

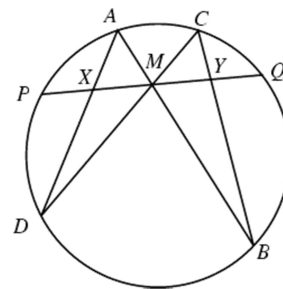
Let ABC be a triangle such that $\angle B$ and $\angle C$ are acute. D is a point on BC such that $AD \perp BC$, E is a point on AC such that $DE \perp AC$. Point M is the midpoint of DE . Show that $AM \perp BE$ if and only if ABC is isosceles.

Problem 5

D and E are points on the sides AB and AC of $\triangle ABC$ such that $BD = CE$. If F and G are midpoints of DE and BC respectively, show that FG is parallel to the angle bisector of $\angle BAC$.

Problem 6 (Butterfly Theorem)

Let M be the midpoint of a chord PQ of a circle, through which two other chords AB and CD are drawn; AD and BC intersect chord PQ at X and Y correspondingly. Prove that M is the midpoint of XY .



Problem 7 (IMO 1995 Problem 1)

Let A, B, C, D be four distinct points on a line, in that order. The circles with diameters AC and BD intersect at X and Y . Let P be a point on the line XY such that $P \notin BC$. The line CP intersects the circle with diameter AC at C and M , and the line BP intersects the circle with diameter BD at B and N . Prove that the lines AM, DN, XY are concurrent.



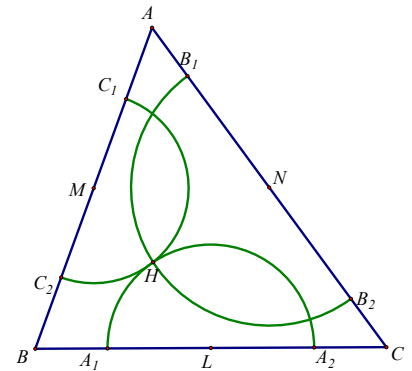
Problem 8 (APMO 2000 Problem 3)

Let ABC be a triangle. Let M and N be the points in which the median and the angle bisector respectively, from A meet the side BC . Let Q and P be the points in which the perpendicular at N to NA meets MA and BA , respectively, and O the point in which the perpendicular at P to BA meets AN produced. Prove that QO is perpendicular to BC .



Problem 9 (IMO 2008 Problem 1)

An acute-angled triangle ABC has orthocenter H . The circle passing through H with centre the midpoint of BC intersects the line BC at A_1 and A_2 . Similarly, the circle passing through H with centre the midpoint of CA intersects the line CA at B_1 and B_2 , and the circle passing through H with centre the midpoint of AB intersect the line AB at C_1 and C_2 . Show that $A_1, A_2, B_1, B_2, C_1, C_2$ lie on a circle.



Problem 10 (IMO 2000 Problem 1)

Two circles G_1 and G_2 intersect at two points M and N . Let AB be the line tangent to these circles at A and B respectively, so that M lies closer to AB than N . Let CD be the line parallel to AB and passing through the point M , with C on G_1 and D on G_2 . Lines AC and BD meet at E ; lines AN and CD meet at P ; lines BN and CD meet at Q . Show that $EP = EQ$.



Problem 11 (IMO 2016 HK Selection Test 4)

Let O be the circumcenter of a triangle ABC , and let l be the line going through the midpoint of the side BC and is perpendicular to the bisector of $\angle BAC$. Determine the value of $\angle BAC$ if the line l goes through the midpoint of the line segment AO .

Problem 12 (CGMO 2021 P2)

As shown in the figure, let ABC be an acute angle with $AB < AC$, point I be its incentre, point J be its ex-centre with respect to A . Points X, Y are on the minor arc AB and minor arc AC of the circumcircle of triangle ABC respectively, such that $\angle AXI = \angle AYJ = 90^\circ$. Point K lies on the extension of BC beyond C such that $KI = KJ$. Show that the line AK bisects the segment XY .

